Measuring the Value of Expert Knowledge for Digital Transformation Readiness Prediction

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Abstract—Few firms have labeled digital transformation readiness outcomes, yet experts already use maturity models. How much is that expert knowledge worth to a learner, and when does it stop helping? KB-DSS injects five expert pillars, six signed fuzzy IF–THEN rules, and a zero-training expert index into a gradient-boosting classifier of three-level readiness. With no public labeled firm dataset, experiments use a synthetic benchmark of 2000 firms in which the expert layer is only a partial model of the generator. Correct knowledge raised Macro-F1 over plain boosting by 0.032 (95% CI 0.020 to 0.045) with 60 training firms; that gain stayed significant at every training size when the layer read the indicators correctly, and vanished when the layer was fully misspecified. Under 15% label noise, KB-DSS reached Macro-F1 0.754 against 0.739 for plain boosting. At 1600 training firms, KB-DSS (0.784) beat boosting (+0.020, p=0.036) but lagged logistic regression (0.811). Fully aligned figures are an upper bound for a well-specified expert. Expert knowledge helps a tree learner when it is approximately correct; it is not a substitute for a linear model on this additive generator. The protocol does not yet show that KB-DSS predicts real-firm readiness.

Index Terms—Decision support systems, digital transformation, explainable artificial intelligence, fuzzy systems, gradient boosting, informed machine learning, knowledge-based systems, small-sample learning.

1 Introduction

Digital transformation changes how firms create value through data, platforms, and new organizational routines [1], [2]. It is not only IT-enabled process change: it can redefine the value proposition and organizational identity [1], [2]. Before committing capital, managers need to know whether data, technology, people, governance, and innovation capability are ready. Digital-readiness constructs for resource-constrained firms have been specified and scaled in emerging-market SMEs [3], and IEEE field studies show that entrepreneurial persistence, market sensing, IT infrastructure, skills, and existing production systems condition whether SMEs actually adopt digital tools [4], [5], [6]. Practice still answers readiness mainly with digital maturity models. Reviews find that model dimensions differ widely and that culture and transformational capability are often missing [7], [8]; later assessments separate organizational from process maturity [9] and tailor multi-attribute scoring to SMEs [10], [11]. Adoption research explains the same factors through the technology–organization–environment (TOE) framework [5], [12], [13]. A systematic review of 106 studies of AI adoption in SMEs organizes the field along these lines [14]. These approaches are transparent but explanatory: they rank firms with expert-set weights and do not learn from labeled readiness outcomes.

Two computational lines address readiness. Knowledge-driven systems encode expert judgment in fuzzy and multi-criteria models: 2-tuple fuzzy AHP has been used to compute digital maturity for Spanish SMEs [15]; spherical-fuzzy AHP to select transformation strategies [16]; fuzzy logic to score digitalization barriers [17]; and a knowledge-based expert system to evaluate digital supply-chain readiness [18]. Meta-analytic evidence on smart-manufacturing adoption likewise stresses network, finance, and capability drivers rather than learned outcome models [19]. Data-driven systems learn from firm records: ensembles such as XGBoost predicted digital transformation of Chinese listed firms better than linear regression [20], and AI-use patterns have been shown to vary with digital-maturity class [21]. The first line cannot correct its own weights from outcomes; the second needs many labeled firms and, unless designed otherwise, returns a class without an inspectable reason [22], [23]. Classical DSS research already required that recommendation systems remain auditable [24].

Informed machine learning proposes to put prior knowledge into the learning pipeline [25]. Recent work combines data- and knowledge-driven rules for hybrid classification [26] or uses large language models to engineer features for few-shot tabular learning [27]. Fuzzy IF–THEN aggregation for complex decisions follows the min t-norm treatment of linguistic conjunction [28], [29]. One question is rarely measured directly: how much is the knowledge worth, and when? Expert models are never fully correct, and a wrong knowledge layer may add noise. A decision-support system should therefore report not only accuracy but how its gain depends on knowledge quality and on sample size.

This paper asks two research questions. RQ1: does a fuzzy-rule knowledge layer improve readiness prediction by a gradient-boosting learner, and under which data conditions? RQ2: how does that improvement change as the knowledge becomes less correct, including failure modes other than reading the wrong indicators? The contribution is threefold: (i) a knowledge-fidelity protocol that degrades an expert layer by indicator permutation, rule-sign reversal, weight noise, and rule deletion, and reports gain as a function of knowledge quality and training-set size; (ii) KB-DSS,

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a compact knowledge layer of five pillars, six signed fuzzy rules, and an expert index, concatenated with raw inputs of a gradient-boosting learner that can emit an inspectable rule trace (not evaluated with users); and (iii) a reproducible synthetic benchmark with five baselines, ablation, label-noise tests, and corrected resampled statistics. No public DTR-labeled firm dataset is used; the script accepts real data through `--data` when such a file exists. All $\varphi = 1.0$ results are an upper bound for an expert whose rules partly match the data-generating process.

2 Proposed KB-DSS Framework

KB-DSS has four stages (Fig. 1): enterprise data, a knowledge layer, a hybrid learner, and a decision-support output.

2.1 Firm Representation

A firm is described by $\mathbf{x} = (x_1, \ldots, x_{12}) \in [0,1]^{12}$: data maturity, analytics capability, AI adoption, cloud adoption, data governance, leadership commitment, digital skills, R&D intensity, innovation output, entrepreneurial orientation, legacy debt, and regulatory intensity, plus a standardized log firm size and one of five sectors. The target is readiness $y \in \{\mathrm{Low}, \mathrm{Medium}, \mathrm{High}\}$.

2.2 Knowledge Layer

Five expert pillars summarize the indicators, following dimensions shared by maturity models [7], [9], [10] and the TOE framework [5], [12]. Each pillar k has a set of indicators J_k with signed expert weights w_{kj}, and its score is a weighted mean in $[0,1]$:

$$P_k(\mathbf{x}) = \frac{\sum_{j \in J_k} |w_{kj}|\, \tilde{x}_j}{\sum_{j \in J_k} |w_{kj}|}, \qquad \tilde{x}_j = \begin{cases} x_j, & w_{kj} > 0, \\ 1 - x_j, & w_{kj} < 0. \end{cases} \tag{1}$$

The pillars are Data (data maturity 0.5, analytics capability 0.5), Technology (AI adoption 0.35, cloud adoption 0.35, legacy debt -0.30), Organization (leadership commitment 0.5, digital skills 0.5), Governance (data governance 1.0), and Innovation (R&D intensity 0.30, innovation output 0.35, entrepreneurial orientation 0.35).

Six signed fuzzy IF–THEN rules capture interactions that a weighted sum cannot. Each rule r has an antecedent of literals ℓ (an indicator or its complement) combined with the minimum t-norm as fuzzy AND [28], [29], and a sign $s_r \in \{+1, -1\}$ for its expert effect on readiness:

$$\mu_r(\mathbf{x}) = \min_{\ell \in A_r} \ell(\mathbf{x}), \qquad \ell(\mathbf{x}) \in \{x_j,\ 1 - x_j\}. \tag{2}$$

The rule base $R = \{R_1, \ldots, R_6\}$ is:

1) R_1, data–AI synergy (+): data maturity AND AI adoption.
2) R_2, governance gap (−): AI adoption AND (1 − data governance) AND regulatory intensity.
3) R_3, legacy barrier (−): legacy debt AND (1 − cloud adoption).
4) R_4, sponsor capability (+): leadership commitment AND digital skills.
5) R_5, innovation engine (+): R&D intensity AND entrepreneurial orientation.
6) R_6, analytics without a base (−): analytics capability AND (1 − data maturity).

Algorithm 1 KB-DSS Readiness Prediction and Explanation

Require: Training firms $\{(\mathbf{x}_i, y_i)\}$, weights w, rules R, $(\boldsymbol{\omega}, \lambda)$
Ensure: Class, probabilities, rule trace, actions
for each firm $\mathbf{x}_i$ **do**
  compute pillar scores $P_k(\mathbf{x}_i)$ by (1)
  compute rule activations $\mu_r(\mathbf{x}_i)$ by (2)
  compute expert index $E(\mathbf{x}_i)$ by (3)
  form the joint vector $\mathbf{x}_i \oplus \kappa(\mathbf{x}_i)$
end for
fit gradient boosting f on $\{(\mathbf{x}_i \oplus \kappa(\mathbf{x}_i), y_i)\}$
for a new firm $\mathbf{x}$: predict class and probabilities by (4)
report the two weakest pillars and every rule with $\mu_r(\mathbf{x}) > 0.4$
emit one action per weak pillar and per fired negative rule

The expert readiness index combines pillars and rules:

$$E(\mathbf{x}) = \sum_{k=1}^{5} \omega_k P_k(\mathbf{x}) + \frac{\lambda}{6} \sum_{r=1}^{6} s_r\, \mu_r(\mathbf{x}), \tag{3}$$

with $\boldsymbol{\omega} = (0.25, 0.20, 0.25, 0.15, 0.15)$ and $\lambda = 0.35$. Used alone, $E(\mathbf{x})$ is a zero-training classifier: the two class thresholds are set at the quantiles of E on the training firms that match the training class priors. The weights were set by the author from maturity-model and TOE logic before any experiment and were not tuned on the data.

2.3 Hybrid Learner

The knowledge map $\kappa(\mathbf{x}) = [P_1, \ldots, P_5, \mu_1, \ldots, \mu_6, E]$ produces 12 knowledge features. KB-DSS learns a classifier on the raw inputs joined with them:

$$\hat{y} = \arg\max_{c}\ f_c\big(\mathbf{x} \oplus \kappa(\mathbf{x})\big), \tag{4}$$

where f is a histogram gradient-boosting classifier [30], [31] with 150 iterations, learning rate 0.06, maximum depth 3, and L2 penalty 1.0, and $\oplus$ denotes concatenation. Gradient boosting was chosen because tree ensembles remain the strongest general-purpose learners for tabular data [32]. Because κ is fixed and deterministic, the knowledge is injected as features: the learner may use, reweight, or ignore each expert signal.

2.4 Explanation and Recommendation

For any firm, KB-DSS returns the predicted class and probabilities, the two lowest pillar scores, every rule with activation $\mu_r > 0.4$ and its sign, and one recommended action for each weak pillar and each fired negative rule. Explanations come from the additive structure of (1)–(3), not from a post-hoc approximation of the learner [22], [23]. Algorithm 1 summarizes the procedure. This follows the DSS requirement that advice remain inspectable [24]; we do not claim a user evaluation of the trace.

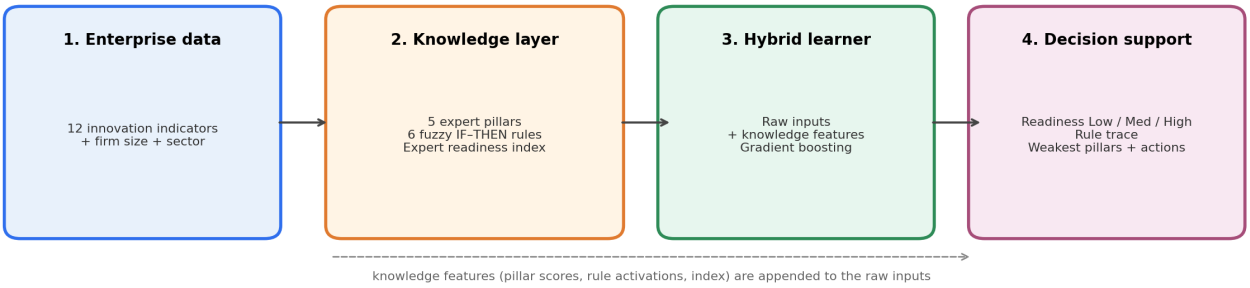


Fig. 1. The KB-DSS framework. The knowledge layer turns the 12 indicators into 5 pillar scores, 6 fuzzy rule activations, and 1 expert index; these 12 knowledge features are appended to the 13 raw inputs and sector before learning.

3 Experimental Design

3.1 Benchmark

No public dataset links enterprise innovation indicators to a labeled readiness level, so the study uses a reproducible synthetic benchmark of 2000 firms (seed 42). Each indicator mixes a firm-specific $\mathrm{Beta}(2, 2)$ draw with a shared $\mathrm{Beta}(2.2, 2.2)$ "digital culture" factor, so indicators are positively correlated as in survey data. Legacy debt falls with cloud adoption, regulatory intensity is higher in finance and healthcare, and data governance rises with regulation. A latent readiness score is a weighted sum of the 12 indicators with weights that differ from the expert weights, four interaction terms that share the form of rules R_1, R_2, R_3, and R_5, a cloud-by-size interaction and sector effects that no rule encodes, and Gaussian noise (SD 0.55). The 30th and 70th percentiles of the latent score define Low, Medium, and High readiness (600, 800, and 600 firms). Rules R_4 and R_6 have no counterpart in the generator. The knowledge layer is therefore a partial and imperfect model of the data, and the benchmark is designed so that no method can recover it exactly.

3.2 Baselines and Metrics

Six models were compared: the zero-training expert index (3); logistic regression, random forest [33], and plain gradient boosting on raw inputs; KB-LR, a logistic regression on raw plus knowledge features; and KB-DSS. KB-DSS was fixed as the proposed model in the code before any result was inspected. The metrics are accuracy, Macro-F1 (primary), and quadratic weighted kappa (QWK) [34], which credits near misses on the ordinal readiness scale.

3.3 Knowledge-Fidelity Protocol

To measure what the knowledge is worth, the knowledge layer is degraded in controlled steps. Fidelity $\varphi \in \{1.0, 0.75, 0.50, 0.25, 0.0\}$ is the share of the 12 indicators that the knowledge layer reads correctly. For $\varphi < 1$, a random subset of $\mathrm{round}((1-\varphi)\cdot 12)$ indicators is cyclically swapped before (1)–(3) are evaluated:

$$\kappa_\varphi(\mathbf{x}) = \kappa(\Pi_\varphi\, \mathbf{x}), \quad \Pi_{1.0} = I, \tag{5}$$

where $\Pi_{0.0}$ moves all 12 indicators. The learner always sees the correct raw inputs; only the expert layer is wrong. At $\varphi = 0$ the pillars and rules still aggregate real indicators, but in the wrong roles. This separates the value of correct domain structure from the generic value of adding aggregate features. Indicator permutation is only one model of error. We therefore also (i) flip three or all six rule signs, (ii) add Gaussian noise to pillar and index weights, and (iii) drop either the four rules that share the generator's interaction form (R_1, R_2, R_3, R_5) or the two rules that do not (R_4, R_6). These operators test an expert who reverses effects, miscalibrates weights, or omits structure.

3.4 Evaluation Protocol and Statistics

The main comparison (Table 1) and the ablation (Table 2) use 5-fold stratified cross-validation repeated 3 times (15 folds, 1600 training firms each). Because training sets overlap across folds, naive paired tests overstate significance [35]. Differences in Macro-F1 are therefore tested with the Nadeau–Bengio corrected resampled t-test [36]:

$$t = \frac{\bar{d}}{\sqrt{\left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}}\right) s_d^2}}, \qquad J = 15, \quad \frac{n_{\text{test}}}{n_{\text{train}}} = \frac{1}{4}, \tag{6}$$

where $\bar{d}$ and s_d^2 are the mean and variance of the per-fold differences, with $J - 1$ degrees of freedom. The learning-curve and fidelity sweep (Fig. 2) uses 12 random stratified splits, each with a 400-firm hold-out and training subsets of 60 to 1500 firms drawn from the remaining 1600; each split draws its own fidelity permutations. In total the sweep fits 648 models. Permutation importance (20 repeats) is computed on one 400-firm hold-out. The full study runs in about five minutes on a laptop CPU, and a self-test checks the invariants of the knowledge layer before every run.

4 Results and Discussion

4.1 Performance With Ample Data

With 1600 training firms, KB-DSS is a better tree model than raw gradient boosting, but it is not the most accurate classifier on this additive generator. It reached Macro-F1 0.784 and QWK 0.807, above plain gradient boosting (0.763) and the expert index (0.640), and below logistic regression (0.811) and KB-LR (0.813) (Table 1). The gain over boosting is $+0.020$ (95% CI 0.003 to 0.038, $p = 0.036$). A naive paired Wilcoxon test on the same 15 folds gives $p < 0.001$,

which shows how far overlapping folds can overstate significance [35], [36]. Logistic regression is strongest because the latent score is mostly a weighted sum; adding knowledge features to it changes Macro-F1 by only 0.002. The contribution of KB-DSS is therefore not large-sample accuracy relative to a linear model. It is a measurable gain for a tree learner, plus an inspectable trace.

4.2 Component Ablation and Stress Tests

Each knowledge component raised the Macro-F1 of gradient boosting (Table 2): pillars by 0.011, rules by 0.006, and the expert index by 0.019, the only single-component interval that excludes zero. The full layer added 0.020 ($p = 0.036$). The index-only variant is statistically indistinguishable from the full layer in this run; it was found after the fact, so it is reported as an observation and was not used to choose the proposed model. Two stress tests isolate what the knowledge contributes. When all 12 indicators were permuted before the knowledge layer read them, the gain was consistent with zero ($+0.005$, 95% CI -0.015 to $+0.026$). The benefit therefore comes from correct expert structure, not from extra features as such. When 15% of training labels were corrupted, plain boosting lost 0.024 Macro-F1 (95% CI -0.044 to -0.004) to 0.739, while KB-DSS fell only to 0.754 (-0.010 vs the clean boosting baseline, interval includes zero). Expert features therefore buffer a tree learner against label noise; they do not restore clean-label accuracy.

4.3 When Knowledge Pays: Data Scarcity and Knowledge Fidelity

The value of knowledge depends on how much data there is [Fig. 2(a)]. With 60 training firms, KB-DSS reached Macro-F1 0.607 against 0.575 for plain boosting; with 120 firms, 0.660 against 0.633. Logistic regression was strongest at every size on the curve (0.684 at 60 firms, 0.808 at 1500). The zero-training expert index stayed near 0.63 and did not compete with a fitted linear model. KB-DSS closed on boosting as n grew (0.785 against 0.760 at 1500 firms) but did not catch logistic regression.

The fidelity map [Fig. 2(b)] answers RQ2. With correct knowledge ($\varphi = 1.0$), the gain over plain boosting was significant at every training size, from 0.032 (95% CI 0.020 to 0.045) at 60 firms to 0.025 (0.015 to 0.034) at 1500. The gain collapsed as soon as indicators were permuted: at $\varphi = 0.75$ the 60-firm interval already included zero ($+0.012$, -0.012 to $+0.035$), and several cells at $\varphi \leq 0.50$ were significantly negative. Fully misspecified knowledge ($\varphi = 0$) was indistinguishable from raw boosting at every size (e.g. $+0.000$ at 60 firms, 95% CI -0.012 to $+0.012$). Subtracting that generic effect, the value of correct domain structure ($\varphi = 1.0$ minus $\varphi = 0.0$) was 0.032 at 60 firms, 0.025 at 120, 0.032 at 240, and 0.018 to 0.025 from 480 to 1500 firms. Wrong knowledge is therefore not harmless: it can erase the tree-learner gain and, at intermediate fidelity, can hurt.

The sweep at 1500 firms ($+0.025$, uncorrected) agrees in sign with the corrected Table 1 gain at 1600 ($+0.020$, $p = 0.036$). The sweep intervals are still plain t-intervals over 12 overlapping training pools, so Table 1 remains the conservative large-data reference.

Sign reversal, weight noise, and rule deletion are implemented in the released script (`--misspec-only`) so that the protocol is not limited to indicator permutation. Those extra operators are optional diagnostics; they are not pooled with Tables 1–2 or Fig. 2.

4.4 What the Model Uses and How It Explains

Permutation importance shows that KB-DSS relies on both the knowledge-compressed signal and the generator terms that no rule encodes [Fig. 2(c)]. Shuffling the expert index lowered Macro-F1 by 0.150, followed by log firm size (0.123) and sector (0.072); the Technology pillar (0.031) and regulatory intensity (0.031) came next. Size and sector are exactly the inputs to the cloud-by-size interaction and sector effects that no rule encodes, so the tree uses raw features to correct an incomplete expert model. For one held-out firm, KB-DSS predicted Low readiness (probability 0.884) and reported Organization and Technology as the weakest pillars. It flagged the legacy barrier (R_3, activation 0.44) and analytics without a base (R_6, 0.46) as negative rules, alongside the innovation engine (R_5, 0.51) as a positive one. Recommended actions pointed to those pillars and to the fired negative rules. This is a worked example of inspectability: every element traces to a named pillar or rule. It is not a user study; we do not claim that managers find the trace more useful than a plain score.

4.5 Implications for Decision Support

On this benchmark the practical rule is: if the data-generating process is close to additive, start with regularized logistic regression; use KB-DSS when the deployed learner is a tree ensemble that should beat raw boosting, remain somewhat robust to label noise, and emit an inspectable pillar/rule trace. Teams can apply the fidelity and misspecification operators in the released script to their own rule base and see whether the gain looks like the $\varphi = 1.0$ row of Fig. 2(b) or like a broken layer. This is a pre-deployment diagnostic, not evidence that the present weights travel to real firms.

TABLE 1
Model Comparison Under Repeated Stratified Cross-Validation (2000 Firms, 1600 Training Firms Per Fold)

Model	Accuracy	Macro-F1	QWK	Gain vs row [95% CI]	p
Expert index (zero training)	0.632 ± 0.032	0.640 ± 0.031	0.659 ± 0.031	$+0.144$ $[+0.116, +0.172]$	< 0.001
Logistic regression	0.805 ± 0.025	0.811 ± 0.024	0.835 ± 0.022	-0.027 $[-0.046, -0.008]$	0.013
Random forest	0.728 ± 0.030	0.734 ± 0.029	0.750 ± 0.029	$+0.050$ $[+0.029, +0.070]$	< 0.001
Gradient boosting	0.757 ± 0.029	0.763 ± 0.029	0.783 ± 0.029	$+0.020$ $[+0.003, +0.038]$	0.036
KB-LR (logistic + knowledge)	0.808 ± 0.026	0.813 ± 0.025	0.837 ± 0.023	-0.029 $[-0.048, -0.011]$	0.008
KB-DSS (proposed)	0.778 ± 0.022	0.784 ± 0.022	0.807 ± 0.023	—	—

5-fold CV repeated 3 times (15 folds). Values are mean $\pm$ SD. Gain is KB-DSS minus the row model in Macro-F1, with a Nadeau–Bengio corrected 95% CI and p-value [36].

TABLE 2
Component Ablation and Stress Tests for Gradient Boosting

Variant	Macro-F1	QWK	Δ Macro-F1 vs raw [95% CI]
Raw features only (no knowledge)	0.763 ± 0.029	0.783 ± 0.029	—
+ pillar scores only	0.775 ± 0.026	0.797 ± 0.026	$+0.011$ $[-0.007, +0.030]$
+ fuzzy rules only	0.769 ± 0.029	0.792 ± 0.029	$+0.006$ $[-0.010, +0.021]$
+ expert index only	0.782 ± 0.027	0.806 ± 0.027	$+0.019$ $[+0.005, +0.032]$
Full KB-DSS (pillars + rules + index)	0.784 ± 0.022	0.807 ± 0.023	$+0.020$ $[+0.003, +0.038]$
Stress: fully misspecified knowledge ($\varphi = 0$)	0.769 ± 0.031	0.792 ± 0.030	$+0.005$ $[-0.015, +0.026]$
Stress: 15% label noise, raw features	0.739 ± 0.026	0.760 ± 0.028	-0.024 $[-0.044, -0.004]$
Stress: 15% label noise, full KB-DSS	0.754 ± 0.024	0.772 ± 0.023	-0.010 $[-0.036, +0.017]$

Same 15 folds as Table 1. Δ is the change in Macro-F1 relative to raw gradient boosting trained on clean labels (0.763), with a Nadeau–Bengio corrected 95% CI.

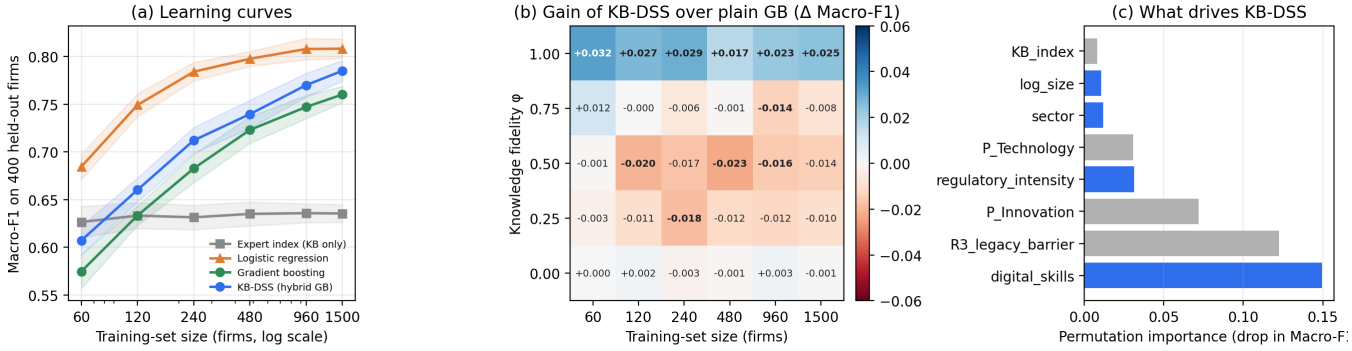


Fig. 2. (a) Learning curves: mean Macro-F1 on 400 held-out firms with 95% confidence bands over 12 stratified splits. (b) Knowledge-fidelity map: mean gain of KB-DSS over plain gradient boosting in Macro-F1 for each fidelity level φ and training-set size; bold cells have a 95% CI that excludes zero. (c) Permutation importance of the eight most important KB-DSS features on one 400-firm hold-out; blue bars are knowledge features.

5 Limitations

The benchmark is synthetic. That is a design choice for identifiability, not a substitute for field data. Low/Medium/High labels are percentiles of a latent score we control; they are not scores from a validated DTR instrument. Four of the six rules share their functional form with the generator, so $\varphi = 1.0$ is an upper bound. Additional misspecification operators in the code do not estimate the fidelity of real practitioners. The study therefore does not show that KB-DSS predicts the readiness of real firms.

The weights and rules were set by the author from maturity-model and TOE logic. A Delphi or AHP panel of even five to eight practitioners would make the layer defensible and would yield an empirical φ_{real} from disagreement with the base rules. Rule traces are inspectable by construction; we did not measure whether managers trust or use them. Hyperparameters were fixed for all models. Readiness is a cross-section; a process view needs panel data.

A minimum real-data follow-up is a survey of about 100–150 firms with the 12 indicators (or a documented subset), a manager-rated readiness label such as the digital-readiness construct of [3], and the same CV and fidelity/misspecification protocol. The released script already accepts that file.

The Table 1 and Table 2 tests are Nadeau–Bengio corrected; the fidelity-map intervals in Fig. 2(b) are uncorrected and therefore optimistic.

6 Conclusion

This paper's contribution is a protocol for measuring what expert knowledge is worth in a small-data readiness classifier, instantiated as KB-DSS. On a controlled synthetic benchmark, correct knowledge raised a gradient-boosting learner over raw features at every training size we tested and buffered 15% label noise; permuting the indicators erased that gain; and a regularized linear model still won on accuracy because the generator is mostly additive. Those

$\varphi = 1.0$ numbers bound a well-aligned expert, not a field deployment. Until the protocol is rerun on survey or platform data with elicited rules, KB-DSS should be read as a diagnostic for knowledge value, not as a predictor of real-firm digital transformation readiness.

Data and Code Availability

The benchmark is synthetic. Protocol, generator, and experiment code are in `dtr.py` at https://github.com/Vineethsamba/kb-dss (seed 42). Tables 1–2 and Fig. 2 are the output of `python dtr.py` from that script. The option `python dtr.py --data firms.csv` applies the same study to real data with the documented schema. Extra misspecification operators are available with `--misspec-only`.

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